

# KABIR THAYANI

## PYTHON & AI DEVELOPER

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### SUMMARY

AI/ML Engineer with hands-on experience training and deploying deep learning models using PyTorch, with a strong focus on computer vision and object detection. Experienced in building CNN-based architectures, fine-tuning YOLO models for real-world detection tasks, and designing end-to-end ML pipelines covering data preprocessing, model training, evaluation, and inference. Proficient in integrating trained models into FastAPI-based production systems, optimizing inference latency, and applying ML to solve practical problems at scale.

### TECHNICAL SKILLS

- AI, ML & Computer Vision: PyTorch, TensorFlow, YOLO, OpenCV, Mediapipe, Scikit-Learn, Deep Learning
- Generative AI & NLP: RAG Pipelines, LLMs, LangChain, Vector Databases, Prompt Engineering
- Frontend & Web: React.js, TailwindCSS, JavaScript, HTML/CSS
- Backend & APIs: FastAPI, RESTful Architectures, n8n Automation
- Data & Storage: PostgreSQL, Pandas, NumPy, Tesseract OCR
- DevOps & Tools: Git, GitHub, Postman

### EXPERIENCE

#### Python Developer Intern

Aarya Softex LLP | June 2025 – August 2025

- Generative AI & RAG: Engineered a Retrieval-Augmented Generation (RAG) chatbot using LangChain and vector embeddings to enable semantic search, improving query accuracy beyond keyword matching.
- Workflow Automation (n8n): Designed and automated end-to-end business workflows using n8n, effectively eliminating manual intervention across multiple internal processes.
- Backend Engineering: Deployed scalable FastAPI services with Role-Based Access Control (RBAC) to ensure secure authentication and permission boundaries for multi-user production workloads.
- AI Agents: Developed an AI-powered voice agent to automate outbound client communication, significantly reducing human involvement in repetitive call workflows.
- Data Pipelines & OCR: Built a real-time Meta Ads lead ingestion pipeline and an OCR + NLP JobScraper (using Tesseract) to extract data from PDF newspapers, improving conversion efficiency by ~70%.

### PROJECTS

#### **CivicLens – AI Pothole Detection & Reporting System**

- Engineered a real-time YOLOv8 inference pipeline to detect road hazards, processing video feeds while automating GPS tagging to eliminate duplicate municipal reports

#### **AI-Powered Applicant Tracking System (RAG & LLM)**

- Engineered a Retrieval-Augmented Generation (RAG) system using FastAPI and ChromaDB to perform semantic search on resumes, identifying candidates based on conceptual fit rather than keywords.
- Deployed a Gemini-based extraction layer to parse unstructured PDFs into JSON, reducing candidate screening time by ~40% and standardizing scoring across diverse resume formats.

#### **AI-Powered Driver Safety Assistant**

- Developed a real-time driver drowsiness detection system using facial landmarks and blink-rate analysis, enabling early alerting for unsafe driving conditions with low-latency inference.
- Optimized computer vision inference pipelines to achieve low-latency, browser-compatible execution, enabling real-time alerting without GPU dependency.

#### **Syllaby – AI Study Planner | React, TailwindCSS, FastAPI, PostgreSQL, Ollama (Gemma 1:2B)**

- Developed an AI-driven study planner generating adaptive schedules, smart flashcards, and progress-tracking dashboards.
- Integrated local LLMs for personalized study material and efficient learning workflows

## Real-Time Face Recognition System

- Engineered an end-to-end computer vision pipeline using OpenCV and Python, implementing Haar Cascade Classifiers for detection and the LBPH algorithm for robust face recognition.
- Developed automated scripts for dataset generation, model training, and real-time inference, incorporating dynamic image preprocessing and a live confidence metric display.

## RAG Chatbot | Flask, LangChain, Vector DB

- Built a Retrieval-Augmented Generation chatbot that answers queries from company documents.
- Implemented vector-based contextual retrieval and Flask API deployment for seamless frontend integration.

## YumYard | MERN Stack (MongoDB, Express, React, Node.js)

- Developed a responsive recipe web app with user personalization and full CRUD operations.
- Implemented authentication and recipe management with persistent MongoDB storage.

## JobScraper | Python, Flask, Tesseract, Gmail API

- Automated job email extraction from Indian e-newspapers using OCR and NLP. Integrated Gmail API (OAuth) for personalized job alerts and designed a scalable backend for data storage.

## n8n Automation | n8n, REST APIs, Workflow Automation

- Created automated workflows for business processes, including email summarization and event triggers.
- Integrated multiple APIs to build fully automated operational pipelines.

## EDUCATION

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### Bachelor of Technology (BTech) in Computer Science Engineering

Noble University | 2026

CGPA: 9.16/10

## CERTIFICATION

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### Full Stack Developer Bootcamp – GeeksforGeeks (2024)

- Mastered frontend (HTML, CSS, JS, React) and backend (Node.js, MongoDB, Mongoose) development.
- Built responsive UIs, implemented APIs, authentication, and MongoDB data handling.

### Introduction to Artificial Intelligence – Simplilearn (2025)

- Studied AI fundamentals: agents, search algorithms, knowledge representation, and reasoning.
- Explored real-world AI applications in chatbots, recommender systems, and computer vision.

### Artificial Intelligence Beginners Guide – Simplilearn (2025)

- Learned ML, DL, and NLP basics with hands-on classification and prediction projects.
- Understood AI project lifecycle from data collection to model deployment.

### Machine Learning using Python – Simplilearn (2025)

- Applied ML algorithms using Python (NumPy, Pandas, Scikit-learn).
- Built predictive models, pipelines, and optimized models via cross-validation.

### Foundation Course on IR4.0 Technologies – Microsoft, SAP & Edunet (2025)

- Explored Industry 4.0 technologies: AI, IoT, Cloud, and Data Analytics.
- Understood automation and smart system integration for digital transformation.

## ADDITIONAL INFORMATION

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- Understanding of Data Structures & Algorithms (DSA) and Object-Oriented Programming (OOP).
- Strong problem-solving skills with focus on performance optimization and scalable application development.
- Actively contributing to open-source projects and maintaining a growing GitHub portfolio.
- Participating in Kaggle competitions to improve ML skills and benchmark model performance.
- Engaged in continuous learning through technical blogs, documentation, and professional certifications.